

Durham Heart Newsletter



Final report for the Heart Hackathon

As the Heart Hackathon competition grows closer, our primary focus has shifted to drafting the final report, which will showcase months of our work. To write the final report we have been gathering and finalising our experimental results. We are collecting critical performance metrics, fluid dynamics data, and structural analysis figures to ensure our submission is backed by solid evidence. Every test that we run and data point we gather allows us to effectively demonstrate the viability of our total artificial heart.



Mechanical Team

Testing and membrane making

We have been conducting physical tests to evaluate our artificial heart, with a particular focus on the membrane. As one of the most critical moving

components, the membrane demands exceptional durability, elasticity, and biocompatibility to withstand continuous pulsatile motion. To evaluate early design concepts, our team developed the initial membrane prototypes completely in-house using layers of liquid latex. This technique allowed us to quickly experiment with varying wall thicknesses and geometries. Future iterations will transition to medical-grade polymers.



Electrical Team

Integrating the TET system

A major milestone involved integrating the wireless TET system and the prototyped artificial heart. The system demonstrated operational efficiency by driving the pumping mechanism to achieve a heart rate of up to 120 RPM while maintaining a low current draw. We aim to minimise heat generation, which is essential for safely powering an implantable device. Our research is also concentrating on how the physical displacement and misalignment between the coils affects the overall energy transfer, which is crucial for ensuring stable, continuous power delivery to a patient.



Business Team

Visit from Dr Chris Bowles

We were honored to host Dr. Chris Bowles, who offered our team an opportunity to present our current artificial heart design, and he provided insightful feedback on our design, clinical considerations, and mechanical performance. We also had the opportunity to see a mock circuit loop directly sourced from a hospital environment. Seeing the mock loop in action showed us key areas where our current experimental setup can be refined.

Thank you!